Beyond Literal Equivalence: An Interpretable System for Terminology-Aware Machine Translation in Historical Texts

Artem Lepin[1,TODO*], **Danil Kostromin**[1,TODO*], **Andrey Sakhovskiy**[1,4], **Semen Budenny**[1,5]
[1]Sber AI [2]TODO [3]TODO [4]Skoltech [5]AIRI

Abstract

Todo: Andrey

1 Introduction

Recent success of Large Language Models (LLMs) have significantly advanced Machine Translation (MT) (Kocmi et al., 2024; Semenov et al., 2025; Feng et al., 2025b; Wang et al., 2025; Tan et al., 2026). However, without explicit grounding, LLMs remain prone to hallucinations and struggle with entity names (Conia et al., 2024b), complex idiomatic expressions (Liu et al., 2023; Kim et al., 2025; Liu et al., 2026) and domain-specific terms (Oncevay et al., 2025).

To assess MT quality, recent research has increasingly adopted the LLM-as-a-judge approach (Kocmi and Federmann, 2023; Kim, 2025; Bavaresco et al., 2025; Feng et al., 2025a; Zhan et al., 2025). Additionally, iterative refinement using feedback from an LLM judge was shown to improve translation quality (Raunak et al., 2023; Wang et al., 2024; Feng et al., 2025b). Despite these advancements, researchers and practitioners lack convenient, interactive tools for in-depth MT error analysis with flexible definition of LLM criteria. Existing evaluation platforms typically rely on rigid, predefined metrics and do not allow users to specify custom, domain-specific evaluation criteria (e.g., evaluating a text specifically for "anachronistic framing" or "ideological neutrality"). Furthermore, they rarely integrate specialized terminology retrieval and normalization pipelines directly into the analysis workflow. This limits the interpretability of the evaluation, as users cannot easily trace whether a translation error stems from a terminology mismatch, a factual hallucination, or a stylistic flaw.

In terminology-rich and culturally nuanced fields, such as history, factual distortions are especially detrimental (**??**). Historical texts present unique challenges: non-matching surface forms in the source language may collapse into identical translations in the target language (e.g., distinct Chinese dynasties such as Qin and Qing mapping to a single Russian term), and region-specific historiographical constructs often lack direct cross-cultural equivalents. Long-standing research direction explores entity- and terminology-aware MT, and retrieval from external sources, to improve cross-cultural adaptation (Zhao et al., 2020a,b; Conia et al., 2024a).

To bridge this gap, we propose SystemName, an interactive web-based tool for the interpretable analysis of MT quality in terminology-rich domain. To allow flexible evaluation, the tool implements a flexible LLM-as-a-judge evaluation framework with arbitrary verbal criteria and refinement suggestions. For terminology analysis, SystemName implement a built-in terminology and normalization module. For improved interpretability, the tool has a Grammarly-like[1] web interface with translation refinement suggestions highlighted as span level and with the reliability categories for translated terms. In summary, this demonstration paper contributes:

- We propose SystemName, an interactive, easily deployable web tool for terminology-aware MT analysis, featuring a novel confidence-based terminology highlighting mechanism that visualizes the resolution of cross-lingual ambiguities via Wikipedia normalization.

- A flexible LLM-as-a-judge evaluation framework that allows users to define, prototype, and apply custom, domain-specific assessment criteria, moving beyond rigid predefined metrics.

*These authors contributed equally to this work.

[1]grammarly.com

- An integrated pipeline that combines translation refinement, entity-aware terminology normalization, and multi-criteria error analysis, providing a comprehensive, interpretable environment for MT researchers and domain experts.

2 System

As shown in Figure **??**, SystemName has three key modules: (i) **Translation**; (ii) **Terminology Extraction**; (iii) **LLM-as-a-judge Evaluation**. All components are integrated into a live web demo that is easily deployable for new languages and with different backbone models.

2.1 Translation

Domain-Specific Translation We generate initial translations using an LLM guided by a specialized prompt. This prompt requires strict preservation of dates, proper names, and historical relations to ensure factual fidelity. It also prioritizes conceptual equivalence over literal translation to avoid syntactic calques from the Russian source. Additionally, the prompt enforces disciplinary standards, including established scholarly terminology, Library of Congress transliteration, BCE/CE dating, and the exclusion of anachronistic framing (full prompt in Appendix 1). To ensure scholarly objectivity, the model was prohibited from introducing anachronistic framing or modern ideological connotations absent from the source text.

Translation Refinement To refine the output, the system uses a multi-criteria evaluation and correction pipeline. First, three separate LLM judges assess the draft translation for Accuracy, Fluency, and Style. Each judge identifies problematic segments and generates structured feedback containing the source fragment, error explanation, and suggested correction. A dedicated refiner LLM then processes the source text, the draft, and the aggregated feedback. This refiner applies the suggested corrections in a single pass, integrating modifications without overlap or contradiction to produce the final translation.

2.2 Terminology Extraction

Terminology Recognition and Normalization. Historical texts require strict preservation of proper names and domain-specific concepts. We extract terminology using a large language model at the paragraph level. The model identifies surface forms corresponding to persons, locations, institutions, and cultural constructs. For each extracted term, the system queries the Wikidata API to retrieve candidate entity identifiers. This search operation grounds the surface form to a unique knowledge graph node. Once linked, the pipeline retrieves the target-language alias or label from the entity's record. This alternative name becomes the normalized translation. Using explicit knowledge graph labels replaces neural generation for proper nouns, reducing factual hallucination. The normalized terms are fed directly into the translation refinement module.

Terminology Disambiguation. Surface-level label matching frequently returns multiple plausible Wikidata candidates, particularly for polysemous historical terms. To resolve such cases, we prompt a second LLM with four inputs: the recognized surface form, the surrounding paragraph context, the list of candidate QIDs, and the natural-language description associated with each candidate. The model is instructed to select the single QID whose description best aligns with the contextual usage, or to abstain when no candidate is defensible. The output is a structured JSON object containing the chosen QID, a confidence score, and a short justification. This design keeps the reasoning traceable and allows downstream modules to treat low-confidence decisions as uncertain rather than silently propagating errors.

Terminology Ambiguity Categories. Based on the disambiguation outcome, every recognized term is assigned to one of three resolution states. **Unambiguous** matches are surface forms that map to a single Wikidata candidate with high label overlap and require no further reasoning. **Context-resolved ambiguities** are cases where multiple candidates are returned and the LLM selects one on the basis of paragraph-level evidence; these are surfaced to the user together with the model's justification. **Unresolved terms** are expressions for which no Wikidata candidate is retrieved, or for which the disambiguation model abstains; the system preserves the source form and flags the span as ungrounded. In the live interface, the three categories are rendered with distinct visual cues so that domain experts can audit the translation at the level of individual terminology decisions.

2.3 LLM-as-a-judge Evaluation

We evaluate translation quality via an LLM-as-a-judge approach that decomposes assessment into three independent dimensions: Accuracy, Fluency, and Style. By isolating these criteria, we prevent error conflation. Each dimension employs a dedicated prompt and rubric, instructing the model to assign a 1–10 score based on error severity and frequency. Additionally, the judges output structured error reports containing the source fragment, problematic translation, error explanation, and suggested correction.

Accuracy This criterion measures semantic fidelity and completeness, treating the source text as ground truth. The judge evaluates factual integrity, proper noun preservation, and terminological accuracy, strictly penalizing omissions, additions, or semantic distortions.

Fluency This criterion assesses the target text's grammatical correctness and naturalness, focusing on the absence of machine translation artifacts and unnatural phrasing. To ensure fair evaluation, the judge uses the source text as a structural reference, avoiding penalties for dense or list-like formatting inherent to the original.

Style Evaluated independently of factual or grammatical correctness, this criterion measures the preservation of the authorial voice, tone, and register. It focuses on rhetorical consistency, penalizing both the stylistic flattening and over-stylization of the source text.

2.4 Live Demo

TODO: Artem: 1. Functionality, 2. Implementation (cite original framework)

2.4.1 Interactive Use Cases

The demonstration interface supports two primary workflows for analyzing and generating domain-specific translations.

Flexible LLM-based Evaluation. Users upload a source text and a candidate translation to receive a detailed quality report. The interface allows users to define or modify LLM-as-a-judge criteria (e.g., factual accuracy, stylistic neutrality) dynamically. The system evaluates the translation against these custom rubrics and outputs a span-level visualization of detected errors. Concurrently, the terminology module extracts domain-specific terms and highlights normalization failures.

End-to-End Translation with Terminology Uncertainty. Users provide a source text alongside custom evaluation criteria. The system executes the translation pipeline, generating an initial draft and refining it to satisfy the specified constraints. The final output presents the refined text with confidence-based terminology highlighting. Terms are visually categorized into three resolution states: (i) **unambiguous** matches grounded directly in Wikidata, (ii) **ambiguities** resolved via context-aware LLM reasoning, and (iii) **unresolved terms** absent from the knowledge base, for which no translation guarantees are provided.

3 Evaluation Setup

3.1 Evaluation Data

3.1.1 WikiHistDataset

Standard machine translation benchmarks lack the terminological density and cross-lingual ambiguity characteristic of historical texts (TODO). Evaluating terminology extraction, entity grounding, and reference-free translation quality in this domain therefore requires specialized resources.

Data Source We construct a custom corpus of 100 articles from the Russian Wikipedia.[2] We sample ten articles from each of ten thematic sections: Sumer and Mesopotamia, Ancient Egypt, Assyria, the Hittite Kingdom, Phoenicia, Achaemenid Iran, Ancient India, Ancient China, Ancient Greece, and Ancient Rome. We select these specific domains because they represent foundational, geographically and culturally distinct cradles of civilization. This selection provides a diverse testbed that maximizes terminological variance. We draw candidates from the corresponding category subtrees using a fixed random seed. A deterministic filter restricts the earliest associated Wikidata date to before 500 CE, ensuring reproducible selection without manual curation (Appendix **??**).

Terminology Annotation For entity annotation, we extract all internal hyperlinks within these articles that point to other Wikipedia pages. We treat the titles of these linked pages as the target terminology entities. Each link target resolves deterministically to a unique Wikidata entity identifier (QID) (Vrandečić and Krötzsch, 2014), providing precise human-annotated ground truth for entity linking without additional labeling cost. How-

[2] https://ru.wikipedia.org

ever, Wikipedia editorial conventions typically restrict entity linking to the first mention of a concept (Kubeša and Straka, 2023). Consequently, spans for repeated entities remain unannotated. Because of this structural limitation, we restrict our evaluation to page-level terminology recognition and normalization rather than exhaustive span-level entity extraction. Our evaluation protocol explicitly accounts for this annotation sparsity (Section **??**). Finally, we utilize these articles as source texts to assess the factual fidelity of machine translation outputs.

BOUQuET (Andrews et al., 2025) is a benchmark for machine translation quality evaluation, with texts manually created by linguists across eight source languages and translated into English. For this study, we use the Russian-to-English subset. The dataset covers eight domains: instructional tutorials, dialogues, narration, social media posts and comments, general web content, reflective pieces, and miscellaneous texts. The test split contains 198 paragraphs (854 sentences, 9,279 words), with an average paragraph length of 47 words.

Russian Historical Encyclopedia. Standard machine translation benchmarks lack the terminological density and strict factual constraints characteristic of academic historiography (TODO). To evaluate terminology-aware translation and entity grounding in this high-stakes domain, we construct a custom corpus from the first volume of the *World History* encyclopedia (Chubarjan (2019), further referred to as *RuWH*), published by the Institute of World History of the Russian Academy of Sciences. This volume (*The Ancient World*) presents extreme challenges for neural machine translation. The source texts exhibit a high entity-to-token ratio, complex cross-lingual terminology mapping, and rigid requirements for factual fidelity, where semantic distortions or anachronistic framing are strictly penalized. For this study, we extract six chapters focusing on i. *Mesopotamia*, ii. *Ancient China*; iii. *Classical Greece*; iv. *Man, Spiritual and Material Culture in the World of Poleis*, v. *Conquests and Empire of Alexander the Great*, vi. *Roman Law*. We select these specific sections to maximize thematic variance and test the system's ability to handle diverse geographical cradles of antiquity and their associated specialized lexicons.

Data Statistics Data statistics for the three corpora are summarized in Table 1.

	BOUQuET	**WikiHist**	**RuWH**
# topics	8 domains	10	6
# words	9,279	160,836	50,231
# paragraphs	198	2,553	542
Avg par.length	47	63	93

Table 1: Evaluation Data Statistics.

3.2 Evaluation Metrics

Terminology Recognition TODO: Artem

Translation Quality To evaluate translation quality, we employ two approaches: (i) LLM-as-a-judge evaluation for *Accuracy*, *Fluency*, and *Style* (Sec. 2.3); (ii) two standard neural metrics: *CometKiwi* (Chimoto and Bassett, 2022) and *MetricX* (Juraska et al., 2024). We adopt these automatic metrics as they are widely adopted in WMT shared tasks and demonstrate strong correlation with human judgments (Andrews et al., 2025; Kocmi et al., 2025).

3.3 Hyperparameter Details

For translation, evaluation, and refinement, all LLM inferences were executed at the paragraph level using the vLLM engine with the following hyperparameters: `temperature` = 0.7, `max_new_tokens` = 20000.

4 Results

4.1 Terminology Recognition

1. TODO: Artem: add terminology table here
2. TODO: Artem: add some findings from the experiments.

TODO: Finding 1: As seen from Table **??**, ...

TODO: Finding 2: As seen from Table **??**, ...

TODO: Finding 3: As seen from Table **??**, ...

4.2 LLM-as-a-judge

1. TODO: Danil: add table with 3+2 metrics here
2. TODO: Danil: add table with 3x2x2 table with correlation coefficients here

4.3 Refiment Improve

3. TODO: Danil: add some findings from the experiments.

TODO: Finding 1: As seen from Table **??**, ...

TODO: Finding 2: As seen from Table **??**, ...

Source Text	Draft (Before)	Refined (After)	Metrics (Before → After)		
			Acc	*Flu*	*Sty*
XXIII . ..	Late 3rd millennium BCE.	End of the 23rd century BCE.	6 → 9	8 → 8	7 → 7
	the Khanir kingdom	the Hana kingdom	5 → 10	7 → 8	7 → 8
(1625–1595 . ..)	Samsu-iluna (1625–1595 BCE)	Samsu-ditana (1625–1595 BCE)	5 → 10	7 → 8	7 → 8
« »	Title of "king of the Primordial Lands"	Title of "king of the Sealand"	5 → 10	7 → 8	7 → 8
	a certain Ur-Nanshe	a certain Urukagina	5 → 10	7 → 8	7 → 8

Table 2: Examples of translation refinement where accuracy was improved. *Acc*, *Flu*, and *Sty* denote the LLM-as-a-judge scores (1–10) for Accuracy, Fluency, and Style, respectively. Arrows indicate the score transition from Before to After refinement.

Model	$R_{\text{doc}}^{\text{term}}$ ↑	P_{label} ↑
Qwen3-4B-Instruct	XX	XX
Gemma-3-27B-it	XX	XX
Qwen3.6-27B	XX	XX
Gemini-3.1-Flash-Lite	0.735 [.724–.745]	*pending*
DeepSeek-V4-Flash†	0.657 [.646–.668]	*pending*
GPT-5.5	XX	XX

Table 3: Terminology extraction and Wikidata grounding on the Wikipedia corpus (100 articles). $R_{\text{doc}}^{\text{term}}$ is document-level recall on the terminology-filtered reference. A gold mention counts as recovered if its entity is predicted anywhere in the same article. The filtered reference excludes structural noise and generic lexical classes that a terminology task should not reward, such as languages and scripts, taxa, materials, and units, leaving 7174 of 7959 human-linked mentions. P_{label} is label-verified precision. Because editors link an entity only at its first mention, the annotation is incomplete, and a prediction absent from the reference is additionally credited when its surface exists as a Wikidata label. The deterministic exact-label path is excluded from this reading, and the value remains a conservative lower bound. The title-sitelink fallback is disabled. Wilson 95% confidence intervals are given in brackets. †4.1% of paragraphs were lost to transient API failures during this run.

TODO: Finding 3: As seen from Table **??**, ...

5 Conclusion

TODO

References

Pierre Andrews, Mikel Artetxe, Mariano Coria Meglioli, Marta R. Costa-jussà, Joe Chuang, David Dale, Mark Duppenthaler, Nathanial Paul Ekberg, Cynthia Gao, Daniel Edward Licht, Jean Maillard, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Eduardo Sánchez, Ioannis Tsiamas, Arina Turkatenko, Albert Ventayol-Boada, and Shireen Yates. 2025. BOUQuET : dataset, benchmark and open initiative for universal quality evaluation in translation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 27515–27535, Suzhou, China. Association for Computational Linguistics.

Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, Andre Martins, Philipp Mondorf, Vera Neplenbroek, Sandro Pezzelle, Barbara Plank, David Schlangen, Alessandro Suglia, Aditya K Surikuchi, Ece Takmaz, and Alberto Testoni. 2025. LLMs instead of human judges? a large scale empirical study across 20 NLP evaluation tasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 238–255, Vienna, Austria. Association for Computational Linguistics.

Everlyn Chimoto and Bruce A. Bassett. 2022. COMET-QE and active learning for low-resource machine translation. In *Findings of the Association for Computational Linguistics: EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, volume EMNLP 2022 of *Findings of ACL*, pages 4735–4740. Association for Computational Linguistics.

A. O. Chubarjan, editor. 2019. *World History*, volume 1. Nauka, Moscow. 2nd ed., revised and expanded. Series prepared by the Institute of World History, Russian Academy of Sciences.

Simone Conia, Daniel Lee, Min Li, Umar Farooq Minhas, and Yunyao Li. 2024a. Enhancing machine translation experiences with multilingual knowledge graphs. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2014, Febru-*

Model	BOUQuET					WikiHist					RuWH				
	Acc	Flu	Sty	MetX	COMET	Acc	Flu	Sty	MetX	COMET	Acc	Flu	Sty	MetX	COMET
Translate Gemma	9.7	9.58	9.53	3.55	0.73	9.47	9.57	8.95	4.43	0.66	9.30	9.33	8.77	4.29	0.61
Translate Gemma Refined	9.91	9.76	9.67	3.58	0.73	9.80	9.65	9.12	4.48	0.66	9.76	9.30	9.00	4.35	0.61
Qwen3.6-27B	9.84	9.92	9.7	3.79	0.72	9.46	9.87	8.80	4.67	0.63	9.18	9.82	8.59	4.72	0.58
Qwen3.6-27B Refined	9.91	9.93	9.76	3.77	0.72	9.82	9.64	9.12	4.66	0.64	9.57	9.45	8.92	4.62	0.60

Table 4: Evaluation results across different datasets. **LLM Judges**: Accuracy (Acc), Fluency (Flu), Style (Sty). TODO: MetX, COMET .

LLM	RuWH		BOUQuET		WikiHist	
	Comet	MetX	Comet	MetX	Comet	MetX
....						
Accuracy	XX	XX	XX	XX	XX	XX
Fluency	XX	XX	XX	XX	XX	XX
Stylistic	XX	XX	XX	XX	XX	XX
Qwen3.6-27B						
Accuracy	XX	XX	XX	XX	XX	XX
Fluency	XX	XX	XX	XX	XX	XX
Stylistic	XX	XX	XX	XX	XX	XX
Qwen3.6-27B Refined						
Accuracy	XX	XX	XX	XX	XX	XX
Fluency	XX	XX	XX	XX	XX	XX
Stylistic	XX	XX	XX	XX	XX	XX

Table 5: Spearman Correlation between LLM and Automatic Evaluations.

ary 20-27, 2024, Vancouver, Canada, pages 23781–23783. AAAI Press.

Simone Conia, Daniel Lee, Min Li, Umar Farooq Minhas, Saloni Potdar, and Yunyao Li. 2024b. Towards cross-cultural machine translation with retrieval-augmented generation from multilingual knowledge graphs. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 16343–16360, Miami, Florida, USA. Association for Computational Linguistics.

Zhaopeng Feng, Jiayuan Su, Jiamei Zheng, Jiahan Ren, Yan Zhang, Jian Wu, Hongwei Wang, and Zuozhu Liu. 2025a. M-MAD: Multidimensional multi-agent debate for advanced machine translation evaluation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7084–7107, Vienna, Austria. Association for Computational Linguistics.

Zhaopeng Feng, Yan Zhang, Hao Li, Bei Wu, Jiayu Liao, Wenqiang Liu, Jun Lang, Yang Feng, Jian Wu, and Zuozhu Liu. 2025b. TEaR: Improving LLM-based machine translation with systematic self-refinement. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 3922–3938, Albuquerque, New Mexico. Association for Computational Linguistics.

Juraj Juraska, Daniel Deutsch, Mara Finkelstein, and Markus Freitag. 2024. Metricx-24: The google submission to the WMT 2024 metrics shared task. In *Proceedings of the Ninth Conference on Machine Translation, WMT 2024, Miami, FL, USA, November 15-16, 2024*, pages 492–504. Association for Computational Linguistics.

Ahrii Kim. 2025. RUBRIC-MQM : Span-level LLM-as-judge in machine translation for high-end models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, pages 147–165, Vienna, Austria. Association for Computational Linguistics.

Jisu Kim, Youngwoo Shin, Uiji Hwang, Jihun Choi, Richeng Xuan, and Taeuk Kim. 2025. Memorization or reasoning? exploring the idiom understanding of LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 21678–21699, Suzhou, China. Association for Computational Linguistics.

Tom Kocmi, Ekaterina Artemova, Eleftherios Avramidis, Rachel Bawden, Ondřej Bojar, Konstantin Dranch, Anton Dvorkovich, Sergey Dukanov, Mark Fishel, Markus Freitag, Thamme Gowda, Roman Grundkiewicz, Barry Haddow, Marzena Karpinska, Philipp Koehn, Howard Lakougna, Jessica Lundin, Christof Monz, Kenton Murray, and 10 others. 2025. Findings of the WMT25 general machine translation shared task: Time to stop evaluating on easy test sets. In *Proceedings of the Tenth Conference on Machine Translation*, pages 355–413, Suzhou, China. Association for Computational Linguistics.

Tom Kocmi, Eleftherios Avramidis, Rachel Bawden, Ondřej Bojar, Anton Dvorkovich, Christian Federmann, Mark Fishel, Markus Freitag, Thamme Gowda, Roman Grundkiewicz, Barry Haddow, Marzena Karpinska, Philipp Koehn, Benjamin Marie, Christof Monz, Kenton Murray, Masaaki Nagata, Martin Popel, Maja Popović, and 3 others. 2024. Findings of the WMT24 general machine translation shared task: The LLM era is here but MT is not solved yet. In *Proceedings of the Ninth Conference on Machine Translation*, pages 1–46, Miami, Florida, USA. Association for Computational Linguistics.

Tom Kocmi and Christian Federmann. 2023. Large language models are state-of-the-art evaluators of translation quality. In *Proceedings of the 24th Annual Conference of the European Association for Machine Translation*, pages 193–203, Tampere, Finland. European Association for Machine Translation.

David Kubeša and Milan Straka. 2023. Damuel: A large multilingual dataset for entity linking. *Preprint*, arXiv:2306.09288.

Emmy Liu, Aditi Chaudhary, and Graham Neubig. 2023. Crossing the threshold: Idiomatic machine translation through retrieval augmentation and loss weighting. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 15095–15111, Singapore. Association for Computational Linguistics.

Linfeng Liu, Saptarshi Ghosh, and Tianyu Jiang. 2026. Evaluating the impact of verbal multiword expressions on machine translation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15291–15319, San Diego, California, United States. Association for Computational Linguistics.

Arturo Oncevay, Elena Kochkina, Keshav Ramani, Toyin Aguda, Simerjot Kaur, and Charese Smiley. 2025. Translating domain-specific terminology in typologically-diverse languages: A study in tax and financial education. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 35030–35044, Suzhou, China. Association for Computational Linguistics.

Vikas Raunak, Amr Sharaf, Yiren Wang, Hany Awadalla, and Arul Menezes. 2023. Leveraging GPT-4 for automatic translation post-editing. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 12009–12024, Singapore. Association for Computational Linguistics.

Kirill Semenov, Xu Huang, Vilém Zouhar, Nathaniel Berger, Dawei Zhu, Arturo Oncevay, and Pinzhen Chen. 2025. Findings of the WMT25 terminology translation task: Terminology is useful especially for good MTs. In *Proceedings of the Tenth Conference on Machine Translation*, pages 554–576, Suzhou, China. Association for Computational Linguistics.

Shaomu Tan, Dawei Zhu, Ke Tran, Michael Denkowski, Sony Trenous, Leonardo F. R. Ribeiro, Bill Byrne, and Felix Hieber. 2026. What does LLM refinement actually improve? a systematic study on document-level literary translation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5929–5957, San Diego, California, United States. Association for Computational Linguistics.

Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: a free collaborative knowledgebase. *Commun. ACM*, 57(10):78–85.

Yutong Wang, Jiali Zeng, Xuebo Liu, Fandong Meng, Jie Zhou, and Min Zhang. 2024. TasTe: Teaching large language models to translate through self-reflection. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6144–6158, Bangkok, Thailand. Association for Computational Linguistics.

Yutong Wang, Jiali Zeng, Xuebo Liu, Derek F. Wong, Fandong Meng, Jie Zhou, and Min Zhang. 2025. Delta: An online document-level translation agent based on multi-level memory. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.

Runzhe Zhan, Zhihong Huang, Xinyi Yang, Lidia S. Chao, Min Yang, and Derek F. Wong. 2025. Are large reasoning models good translation evaluators? analysis and performance boost. *CoRR*, abs/2510.20780.

Yang Zhao, Lu Xiang, Junnan Zhu, Jiajun Zhang, Yu Zhou, and Chengqing Zong. 2020a. Knowledge graph enhanced neural machine translation via multi-task learning on sub-entity granularity. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 4495–4505, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Yang Zhao, Jiajun Zhang, Yu Zhou, and Chengqing Zong. 2020b. Knowledge graphs enhanced neural machine translation. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI 2020*, pages 4039–4045. ijcai.org.

A Example Appendix

This is an appendix.

B LLM prompts

```
% \footnotesize
"# Stage 2     Academic English Draft

## Role and context

You are an expert academic translator specializing in historical, archaeological,
    and historiographical literature, working in the tradition of *Encyclopedia
    Britannica* and *Cambridge Ancient History*. Your task is to translate passages
    of a Russian-language historical encyclopedia into scholarly English that meets
    international academic standards.\n

## Mandate

- **Factual fidelity.** Preserve every fact in each paragraph: dates, numbers,
    proper names, asserted relations, and modalities. Do not add facts, omit facts,
    or move facts across paragraphs.
- **Sentence restructuring.** You may split one Russian sentence into several
    English ones, or combine several into one, whenever that produces clearer or
    more natural English. Sentence-level merging and splitting *inside* a paragraph
    is expected and encouraged.
- **No calques.** Do not transpose Russian sentence structure. Build each English
    sentence from the facts it contains, as if writing the same material originally
    in English.
- **Conceptual equivalence over literal translation.** Prioritize meaning over word-
    for-word rendering; never use a literal equivalent that misrepresents historical
     reality.
- **Human scholarly prose.** The output must read as prose a specialist human
    scholar would write     not as a translation, and not as machine-generated text.

## 1. Terminological and historiographical equivalence

- Identify all domain-specific terms, named entities, institutions, and culture-
    bound concepts. Never translate them literally if the direct equivalent
    misrepresents historical reality.
- Apply the established English academic equivalent used in peer-reviewed literature
     (e.g., archaeology, ancient Near Eastern studies, historiography).
- When no direct equivalent exists, use a descriptive translation with minimal
    contextual clarification on first mention only     e.g., *nomos-based
    administrative units (early Sumerian territorial divisions)*     then use the
    standardized form thereafter.
- Maintain strict terminological consistency throughout the entire output. Once you
    choose an English rendering for a Russian term or proper noun, use that
    rendering without variation.
- When Russian historiography employs region-specific conceptual frameworks (e.g., *
                                             *, *
                    *, *                                   *), translate them into
    their recognized English academic counterparts or render them neutrally with
    brief contextualization.

## 2. Proper nouns and transliteration

- Follow standard academic transliteration conventions     Library of Congress for
    Russian, or the field-specific norm for archaeological and ancient terms     for
     all non-Latin names.
- On first occurrence, introduce a transliterated term with a brief English
    explanation, then use the standardized English form or accepted short form
    thereafter.
- Retain widely recognized English exonyms unchanged: *Mesopotamia*, *Uruk*, *Jemdet
     Nasr*, *Indus Valley*, etc.

## 3. Chronology and dating conventions

- Standardize all dates to **BCE/CE** format unless the source explicitly requires
    BC/AD for stylistic consistency.
- Format periods consistently: *4th millennium BCE*, *mid-3rd century BCE*, *Late
    Bronze Age*.
- If non-Gregorian or culture-specific dating systems appear in the source, clarify
    them without altering the source's chronological claims.

## 4. Cultural and conceptual adaptation

- Avoid anachronistic framing or projecting modern categories onto ancient phenomena
     .
- Do not introduce modern ideological or political connotations absent from the
    source.
- Preserve the authoritative, reference-style tone characteristic of major academic
```

LLM system prompt for facts extraction.

*You are an **expert fact extraction** and **verification assistant**.*

Please read the following text carefully and break it down into distinct, independent facts. For each fact, disambiguate it to ensure clarity and precision (e.g., replace ambiguous prepositions).

Each fact should be written on its own line. Each line must start with a hyphen and space ('- '). Do not include any additional explanation or formatting - just the list of facts if there are any.

LLM system prompt for facts verification.

*You are an **expert fact extraction** and **verification assistant**.*

*You are given an atomic fact and a list of textual passages. Your task is to determine whether the atomic fact is **True** or **False** based solely on the information in the passages.*

Instructions:

1. *Check if any of the passages directly support the atomic fact.*
2. *Output **'True'** if at least one passage supports the fact even if another passage contradicts the fact.*
3. *Output **'False'** if no passage supports the fact.*
4. *Do not include any additional information and explanations, you must only answer **'True'** or **'False'**.*